

Introduction

If you are a farmer, landowner, farmhouse owner, agricultural investor, or someone planning a water project in Pakistan, one question comes first:

What is the actual solar tube well price in Pakistan?

The simple answer is: a solar tube well in Pakistan can cost anywhere from PKR 8 lakh to PKR 40 lakh+, depending on pump horsepower, bore depth, water requirement, solar panel capacity, VFD quality, installation structure, pipe length, cable size, and site conditions.

But that answer alone is not enough.

Because the real question is not only “What is the price?” The real question is:

Which solar tube well system is right for your land, your water depth, your crop, your budget, and your long-term farming cost?

Many people make the mistake of buying a solar tube well system by only asking for the lowest quotation. Later, they face low water flow, pump failure, weak structure, VFD problems, wrong panel sizing, poor wiring, or a system that runs only for a few hours properly.

A solar tube well is not just a solar panel setup. It is a complete water-lifting project. It includes solar panels, pump, motor, VFD/controller, structure, cables, protection system, delivery pipeline, bore condition, water depth, discharge requirement, and installation quality.

This guide is written to help you understand the real price of solar tube wells in Pakistan, what affects the cost, what system size you need, and how to make a smart decision before spending your money.

Quick Price Overview

The price of a solar tube well depends mainly on pump horsepower and water depth. Below is a practical market-based estimate for complete solar tube well systems in Pakistan.

Pump Size Solar Capacity Usually Required Estimated Price Range Suitable For
3HP Solar Tube Well 3kW – 5kW PKR 6 lakh – 9 lakh Small farms, shallow bore, low water demand
5HP Solar Tube Well 5kW – 7kW PKR 8 lakh – 12 lakh Small to medium farms
7.5HP Solar Tube Well 8kW – 11kW PKR 12 lakh – 18 lakh Medium farms, better water flow
10HP Solar Tube Well 10kW – 14kW PKR 15 lakh – 22 lakh Medium to large farms
15HP Solar Tube Well 15kW – 20kW PKR 22 lakh – 30 lakh Large farms, deeper bore, higher discharge
20HP Solar Tube Well 20kW – 28kW PKR 28 lakh – 40 lakh Large agricultural land
30HP+ Solar Tube Well 30kW+ PKR 40 lakh – 60 lakh+ Commercial agriculture and high-volume irrigation

Important: These prices are not fixed. A 10HP system at one site may cost PKR 16 lakh, while another 10HP system may cost PKR 22 lakh because the bore depth, water level, pipe length, panel brand, VFD brand, pump quality, and installation work can be completely different.

What Is a Solar Tube Well?

A solar tube well is a water pumping system that uses solar energy to run a pump and extract water from underground sources. Instead of using diesel engines or grid electricity, the system uses sunlight to power the pump.

A typical solar tube well system includes:

Component	Purpose
Solar Panels	Generate electricity from sunlight
VFD / Solar Pump Controller	Controls power supply and pump speed
Submersible Pump / Turbine Pump	Lifts water from underground
Motor	Drives the pump
Mounting Structure	Holds solar panels safely
Cables and Wiring	Transfers power from panels to controller and pump
Protection System	Protects against voltage issues, dry run, overload, and faults
Delivery Pipe	Carries water to field channel, tank, or irrigation system

A good solar tube well is designed according to your water depth and irrigation requirement. A poor system is designed only according to your budget. That is where most problems begin.

Why Solar Tube Wells Are Becoming Popular in Pakistan

Solar tube wells are becoming popular because farmers are tired of diesel cost, electricity bills, load shedding, transformer issues, and unpredictable operating expenses.

For many farmers, the monthly running cost of a diesel tube well can become a serious burden. Diesel prices change, engine maintenance continues, oil changes are needed, and breakdowns happen during important crop stages. Electricity is also not always reliable, especially in rural agricultural areas.

Solar solves one big problem: running cost.

Once installed properly, a solar tube well can run during daylight hours with very low operating cost. There is no diesel purchase, no electricity unit bill for pumping, and no daily fuel arrangement.

But solar also needs responsible water use. Since the running cost becomes very low, some farmers may over-pump water. That can damage the water table over time. A smart farmer should not only install solar but also manage water properly through scheduling, storage, pipeline, drip, sprinkler, or center pivot irrigation where suitable.

Solar Tube Well Price by HP

3HP Solar Tube Well Price in Pakistan

A 3HP solar tube well usually costs around PKR 6 lakh to 9 lakh, depending on whether the system is basic or complete.

This system is suitable for:

- Small farms
- Shallow water level
- Low daily water demand
- Kitchen farming
- Small orchards
- Farmhouses

- Livestock water supply

A 3HP system is not ideal for large-scale flood irrigation. If your water depth is high or your land is large, choosing 3HP only to save money can create problems later.

5HP Solar Tube Well Price in Pakistan

A 5HP solar tube well usually costs around PKR 8 lakh to 12 lakh.

This is one of the most common choices for farmers with small to medium agricultural land. It offers better water flow than 3HP and can work well where the bore depth and water requirement are moderate.

Best for:

- 3 to 7 acres depending on crop and water level
- Medium water flow requirement
- Existing bore conversion
- Diesel tube well replacement
- Small orchards and vegetable farms

Before installing a 5HP system, check your actual water level, bore diameter, pump depth, and required daily irrigation hours.

7.5HP Solar Tube Well Price in Pakistan

A 7.5HP solar tube well usually costs around PKR 12 lakh to 18 lakh.

This size is useful when 5HP is not enough but 10HP is more than required. Many farmers with medium land and better irrigation demand choose this range.

Best for:

- Medium farms
- Deeper bore than usual
- Better water discharge

- Existing 7.5HP electric or diesel setup
- Farms where water flow matters more than minimum price

This system usually needs a stronger VFD/controller, proper wiring, and a good panel structure.

10HP Solar Tube Well Price in Pakistan

A 10HP solar tube well usually costs around PKR 15 lakh to 22 lakh.

This is a serious agricultural system. A 10HP system is commonly used for medium to large farms, especially where water demand is higher and the farmer wants a reliable daily water supply.

Best for:

- Medium to large farms
- Existing diesel tube well conversion
- Stronger water discharge
- Deeper bore applications
- Larger irrigation channels
- Farms growing crops with higher water demand

A 10HP system must not be designed casually. The VFD, pump, panels, structure, and cable sizing should be properly calculated. Wrong design can reduce water output and damage the pump.

15HP Solar Tube Well Price in Pakistan

A 15HP solar tube well usually costs around PKR 22 lakh to 30 lakh.

This is for larger agricultural operations where water demand is high. It is also used when the water table is deeper or when the farm needs higher discharge.

Best for:

- Large farms

- Deep bore
- High water requirement
- Large crop area
- Commercial agriculture
- Long delivery pipeline

For 15HP systems, quality becomes extremely important. Saving money by using weak equipment can create major losses. If the VFD fails, the pump burns, or panels underperform, the whole irrigation cycle can be disturbed.

20HP and Above Solar Tube Well Price in Pakistan

A 20HP system can cost PKR 28 lakh to 40 lakh, while 30HP+ systems can go above PKR 40 lakh to 60 lakh.

These systems are usually for:

- Large landowners
- Commercial agriculture
- High-volume water supply
- Deep bore
- Heavy irrigation demand
- Large pipeline systems
- Corporate farms

These projects require proper engineering, not guesswork. Site survey, water level checking, flow requirement, cable calculations, structure design, and safety protection are all necessary.

What Affects Solar Tube Well Price?

The price of a solar tube well is not based on HP alone. Two farmers may both ask for 10HP, but one may need a much more expensive system because his water level is

deeper or his field is far from the bore.

Here are the main factors.

1. Bore Depth and Water Level

Bore depth is one of the biggest cost factors.

A shallow bore requires less pumping force. A deeper bore requires more head, more pump power, better pipe strength, and sometimes a larger solar system.

Water Condition Effect on Price Shallow water level Lower cost Medium depth Moderate cost Deep water level Higher cost Falling water table Needs stronger planning Poor bore condition May need repair or new bore

Before buying a solar system, always check:

- Total bore depth
- Static water level
- Dynamic water level
- Bore diameter
- Existing pump depth
- Water discharge
- Sand issue
- Bore age and condition

2. Pump Size and Type

The pump is the heart of the system. If the pump is too small, water flow will be weak. If it is too large, the system becomes expensive and may not run efficiently.

Common pump types include:

- Submersible pump
- Turbine pump

- Surface pump
- Solar DC pump
- AC pump with VFD

For agricultural tube wells, AC submersible or turbine systems with VFD are commonly used for larger setups. DC pumps may work well for smaller systems, but for bigger agricultural projects, AC systems with proper control are often preferred.

3. Solar Panel Capacity

Solar panels decide how much power your system can generate. If panels are under-sized, your pump will not run at proper capacity. It may start late, stop early, or give weak water flow.

For example:

Pump Size Common Solar Capacity 3HP 3kW – 5kW 5HP 5kW – 7kW 7.5HP 8kW – 11kW
10HP 10kW – 14kW 15HP 15kW – 20kW 20HP 20kW – 28kW

Panel quality also matters. A cheaper panel may reduce initial price, but a reliable panel gives better long-term performance.

4. VFD or Solar Pump Controller

The VFD/controller controls the pump. It adjusts power, frequency, and motor operation according to sunlight. This is one of the most important parts of the system.

A low-quality VFD can cause:

- Pump tripping
- Low water output
- Motor heating
- System shutdown
- Frequent faults
- Shorter equipment life

A good VFD should have:

- MPPT function
- Dry-run protection
- Overload protection
- Under-voltage protection
- Motor safety features
- Proper cooling
- Field reliability
- Correct sizing according to pump

Do not treat the VFD as a small accessory. In a solar tube well, it is a major control unit.

5. Structure Quality

The structure holds solar panels in open agricultural land. It must handle wind, dust, heat, and long-term outdoor exposure.

A weak structure can bend, shake, rust, or damage panels. A good structure costs more, but protects your investment.

Check:

- Material thickness
- Galvanized or painted finish
- Wind resistance
- Foundation strength
- Tilt angle
- Panel spacing
- Height from ground
- Theft/security planning

6. Cable and Wiring

Cable quality is often ignored, but it affects safety and performance. Long cable runs need proper sizing. Thin or poor-quality cable causes voltage drop, heating, energy loss, and safety risks.

Good wiring should include:

- Correct DC cable size
- Correct AC cable size
- Proper earthing
- Cable protection
- UV-resistant outdoor cable
- Safe joints
- Proper conduit or trenching where needed

7. Installation Site

Every site is different. Installation cost changes according to:

- Distance from city
- Ground condition
- Foundation requirement
- Distance between panels and bore
- Pipe length
- Cable length
- Accessibility for machinery
- Security needs
- Existing bore condition

A system installed near Multan city may have different logistics cost than a system

installed far away in a rural area.

8. New Installation vs Conversion

There are two cases:

Case Cost Impact Existing tube well conversion to solar Usually lower New bore + pump + solar system Higher Diesel to solar conversion Medium to high Old pump replacement with solar setup Depends on pump condition Complete water project from zero Highest

If your bore and pump already exist and are in good condition, conversion may be easier. But if the bore is weak, pump is old, or water output is poor, only adding solar will not solve the real problem.

Complete Solar Tube Well Cost Breakdown

Here is a practical cost structure.

Component Share in Total Cost Notes Solar Panels 35% – 45% Depends on kW size and panel brand Pump and Motor 15% – 25% Depends on HP, depth, and brand VFD / Controller 8% – 15% Critical for system performance Structure 8% – 15% Depends on strength and design Cables and Protection 5% – 10% Depends on distance and system size Pipes and Fittings 5% – 10% Depends on delivery distance Installation and Commissioning 5% – 10% Depends on site and complexity

The cheapest quotation usually cuts cost from one of these areas. That may look attractive in the beginning, but it can become expensive later.

Solar Tube Well vs Diesel Tube Well

Many farmers compare solar with diesel because diesel tube wells have been used for years. Diesel has one advantage: it can run at night and anytime if fuel is available. But the running cost is high.

Factor	Solar Tube Well	Diesel Tube Well
Initial Cost	High	Lower
Running Cost	Very low	Very high
Fuel Requirement	No fuel	Diesel required daily
Maintenance	Low to medium	High
Noise	Low	High
Operation Time	Daylight hours	Anytime with fuel
Long-term Cost		

Lower Higher Best For Long-term farming Temporary or backup use

Which One Saves More Money?

Solar usually wins in long-term cost. Diesel may look cheaper at the start, but fuel cost keeps increasing. If you use your tube well regularly, solar can recover its cost over time through fuel savings.

Solar is especially useful for farmers who:

- Run tube wells daily
- Spend heavily on diesel
- Face electricity issues
- Want predictable irrigation cost
- Plan to farm long-term
- Need water during daylight hours

Diesel may still be useful as backup for night irrigation or emergency use.

Solar Tube Well vs Electric Tube Well

Electric tube wells can work well where electricity supply is stable and tariff is manageable. But many rural areas face issues like load shedding, transformer problems, voltage fluctuation, and billing pressure.

Factor	Solar Tube Well	Electric Tube Well	Dependency	Sunlight	Grid electricity
Monthly Bill	Very low	High depending on usage	Load Shedding Risk	No	grid
dependency	Yes	Voltage Issues	Controlled through VFD	Common in rural areas	Initial
Cost	High	Lower if grid exists	Long-term Cost	Lower	Depends on tariff

Solar is not always the cheapest on day one, but it is often better for long-term control.

Which Solar Tube Well Size Do You Need?

The right size depends on five things:

- Water depth
- Required water flow
- Land area
- Crop type
- Daily irrigation hours

Do not select HP only by acres. Two farms with the same acreage may need different systems because water depth and crop requirement may differ.

Basic Selection Table

Farm Situation	Recommended System
Small farm, shallow bore	3HP – 5HP
Medium farm, normal water depth	5HP – 10HP
Medium farm, deeper bore	7.5HP – 15HP
Large farm, high water demand	15HP – 20HP
Commercial agriculture	20HP+
Orchard with controlled irrigation	Depends on flow and irrigation method
Drip or sprinkler system	
Needs pressure-based design	Center pivot irrigation
	Requires engineered system

Do Not Oversize Blindly

Bigger is not always better. An oversized system means:

- Higher initial cost
- More panels
- Bigger structure
- Higher cable cost
- More protection requirement
- Possible wastage of water

Do Not Undersize to Save Money

An undersized system means:

- Low water flow

- Poor irrigation
- Longer running hours needed
- Pump stress
- Farmer dissatisfaction
- Possible crop loss

The best system is the one that matches your actual site.

How Many Solar Panels Are Needed?

The number of solar panels depends on panel wattage and system size.

For example, if you use 580W panels:

System Size Approx. Panels Required
 5kW 9 panels 7kW 12 panels 10kW 18 panels
 14kW 24 panels 20kW 35 panels 28kW 49 panels

This is only an estimate. Final design depends on VFD voltage range, pump load, panel wattage, string design, sunlight conditions, and safety margins.

AC or DC Solar Pump: Which Is Better?

Both AC and DC solar pumps have their place.

Feature	AC Solar Pump	DC Solar Pump	Best For
Medium and large agricultural systems			
Small systems	Controller VFD required	DC controller required	Availability
Widely available	Available but more specific	Repair Support	Easier in many areas
Depends on brand	Cost Medium to high	Low to medium	Large HP Use
Better	Limited	Long-term	Flexibility
Good	Limited		

For small applications, DC pumps can work well. For agricultural tube wells, especially 5HP and above, many farmers prefer AC pumps with VFD because service, replacement, and scaling options are often more practical.

Should You Add Batteries?

For most agricultural solar tube wells, batteries are not recommended.

Why?

Because tube wells require high power. Battery backup for a large pump becomes very expensive. It also increases maintenance and replacement cost.

A better approach is:

- Run pump during sunlight
- Store water in tank or pond if needed
- Use efficient irrigation scheduling
- Use pipeline or drip system to reduce wastage
- Keep diesel or grid as backup only if necessary

Batteries may make sense for small domestic water systems, but for large agricultural pumping, they usually make the project unnecessarily expensive.

Common Mistakes Farmers Make

Mistake 1: Buying Only by Lowest Price

The lowest quote may use weak panels, poor VFD, undersized wiring, low-quality pump, or weak structure.

A solar tube well is a long-term project. Saving 1–2 lakh today can cause losses later if the pump fails during crop season.

Mistake 2: Not Checking Water Depth

A quotation without water depth is incomplete. You cannot size a pump properly without knowing the water level.

Mistake 3: Ignoring Bore Condition

If the bore itself is weak, old, sandy, or low-yielding, solar will not magically fix the water issue.

Mistake 4: Weak Structure

Solar panels stay in open fields for years. A poor structure can damage the entire investment.

Mistake 5: Wrong VFD Selection

The VFD must match the pump and solar array. Wrong VFD sizing can cause frequent faults.

Mistake 6: No Protection System

A proper system needs protection against dry run, overload, short circuit, voltage issues, and earthing problems.

Mistake 7: No After-Sales Support

Installation is only the first step. You need a team that can support you after installation.

Before You Buy: Site Survey Checklist

Before finalizing a solar tube well system, collect this information:

Question Why It Matters What is total bore depth? Helps calculate pump head What is static water level? Shows resting water depth What is dynamic water level? Shows water level while pumping What is bore diameter? Decides pump compatibility What is current pump HP? Helps conversion planning What is required water flow? Decides pump size What crop is grown? Decides irrigation demand How many acres need irrigation? Helps estimate system size Is there shading near the site? Affects solar output How far is field from bore? Affects pipe and cable cost Is water sandy? Affects pump selection Is theft/security a concern? Affects structure planning

A professional company should ask these questions before giving a serious quotation.

Payback Period: When Does Solar Recover Its Cost?

The payback period depends on how much diesel or electricity cost you are currently spending.

Here is a simple example.

Item	Diesel	Tube Well	Solar	Tube Well	Initial Setup	Lower	Higher	Monthly
Fuel/Electricity	High	Very low	Maintenance	High	Low to medium	3-Year	Running	Cost
High	Low	Long-Term	Savings	Lower	Higher			

If a farmer spends PKR 40,000 to PKR 100,000 per month on diesel and maintenance, the annual cost becomes very high. In such cases, solar can recover its cost in a few years, depending on usage.

The more you use your tube well, the faster solar makes financial sense.

Government Solar Tube Well Subsidy in Pakistan

Government schemes for solar tube wells have been announced from time to time, especially in Punjab. These schemes usually aim to reduce farmers' energy cost and promote solar-powered irrigation.

However, farmers should be careful. Subsidy programs can have:

- Eligibility conditions
- Application deadlines
- Documentation requirements
- Land ownership requirements
- Existing tube well requirements
- Capacity limits
- Approval delays
- Vendor selection rules

So do not make your whole plan depend only on subsidy. If subsidy is available, it is a benefit. But your system should still be selected based on technical need, water requirement, and long-term reliability.

Before applying, check:

- CNIC requirement
- Land ownership documents
- Tube well registration
- Electricity/diesel tube well proof
- Agriculture department portal
- Last date
- Approved vendor list
- Required contribution from farmer
- System capacity limits

Is Solar Tube Well Good for Every Farmer?

Solar tube wells are excellent for many farmers, but not for everyone.

Solar Tube Well Is Good If:

- You have regular irrigation needs
- Diesel cost is too high
- Electricity supply is unreliable
- Your bore has good water
- You farm during daylight irrigation hours
- You plan to use the land long-term
- You want stable operating cost
- You can invest upfront

Solar Tube Well May Not Be Right If:

- Your bore has very low water

- You need mostly night irrigation
- Your land is rented short-term
- Your water table is falling badly
- You cannot secure panels from theft
- Your usage is very low
- You only want the cheapest possible system

A good company should tell you honestly whether solar is suitable for your land or not.

Smart Water Management with Solar Tube Wells

Solar makes pumping cheaper, but water is still limited. If a farmer starts pumping carelessly just because sunlight is free, the land may face water problems later.

To use solar responsibly:

- Avoid unnecessary over-pumping
- Use water according to crop need
- Repair field channels
- Use pipeline where possible
- Consider drip or sprinkler irrigation
- Use water storage if needed
- Monitor bore water flow
- Check water quality
- Schedule irrigation properly

Solar should not only reduce cost. It should improve farm water management.

Best Crops for Solar Tube Well Use

Solar tube wells can support many crops, but water planning depends on crop type.

Crop / Use Solar Tube Well Suitability Wheat Good Cotton Good with proper scheduling
Mango orchard Good Citrus orchard Good Vegetables Very good Fodder Good but water
demand can be high Rice Needs careful water planning Livestock water Very good
Farmhouse water Good Drip irrigation Excellent

For water-heavy crops, the system must be designed carefully to avoid under-sizing.

New Bore + Solar Tube Well vs Existing Bore Conversion

Existing Bore Conversion

If you already have a working bore and pump, solar conversion may be easier. The company checks your current pump size, water flow, motor type, voltage, bore depth, and energy requirement. Then it designs a solar system and VFD to run the pump.

Pros:

- Lower cost than complete new project
- Faster installation
- Uses existing bore
- Good for diesel/electric replacement

Cons:

- Existing pump may not be suitable
- Old motor may fail
- Bore may have low yield
- Existing pipes may need replacement

New Bore + Solar System

This is a complete water project from zero. It includes bore drilling, pump selection, solar sizing, pipework, and installation.

Pros:

- Properly designed from start
- Better long-term planning
- Pump and solar can be matched correctly
- Good for new farms

Cons:

- Higher cost
- Needs water investigation
- More time required
- Bore success depends on underground water conditions

What Should Be Included in a Professional Quotation?

A proper quotation should not just say “10HP Solar Tube Well — 18 lakh.”

It should mention:

Quotation Item Should Be Clear? Pump HP Yes Pump brand/type Yes Motor rating Yes
Solar panel brand and wattage Yes Total kW capacity Yes VFD/controller brand and
rating Yes Structure type Yes Cable size and length Yes Protection system Yes
Pipe/fitting details Yes Installation scope Yes Warranty Yes After-sales support Yes
Timeline Yes Payment terms Yes

If the quotation is vague, the project can become risky.

Recommended System by Farmer Type

Farmer Type Suggested Approach Small farmer 3HP-5HP, avoid overspending Medium
farmer 5HP-10HP, focus on correct sizing Large farmer 10HP-20HP, professional
design required Orchard owner Flow + pressure-based design Commercial farmer
Engineered solution with monitoring Farmhouse owner Small solar pump or hybrid
system Diesel user Compare monthly diesel cost with solar EMI/payback New

landowner Start with water investigation and bore planning

Why FJ Group's Approach Is Different

FJ Group treats solar tube well installation as a complete water project, not just a solar product sale.

A good project starts before installation. It starts with understanding your land, water depth, crop requirement, current pumping system, and future irrigation plan.

FJ Group can guide you through:

- Water requirement assessment
- Bore and water condition review
- Solar tube well sizing
- Pump and VFD selection
- Diesel to solar conversion
- New solar tube well project planning
- Irrigation and water management planning
- Industrial and agricultural water solutions

This matters because your goal is not just to install panels. Your goal is to get reliable water for your farm at the right cost.

Final Buying Guide: Which System Should You Choose?

Use this simple decision guide.

Your Situation Best Decision You have 3-5 acres and shallow water 3HP-5HP system
You have 5-10 acres and normal water depth 5HP-10HP system You have deep water
level Do survey before selecting HP You already use diesel tube well Calculate diesel
cost and convert to solar You have old bore with low flow Fix bore issue before solar
You need high pressure irrigation Design pump according to pressure You want
drip/sprinkler/center pivot Need engineered solution You want lowest price only High

risk You want long-term reliability Choose proper design and quality equipment

Frequently Asked Questions

1. What is the price of a solar tube well in Pakistan?

The price usually starts from around PKR 8 lakh for smaller practical systems and can go above PKR 40 lakh for large agricultural systems. The final price depends on HP, depth, solar capacity, pump, VFD, structure, cables, and installation.

2. What is the price of a 5HP solar tube well in Pakistan?

A 5HP solar tube well usually costs around PKR 8 lakh to 12 lakh, depending on equipment quality and site conditions.

3. What is the price of a 10HP solar tube well in Pakistan?

A 10HP solar tube well usually costs around PKR 15 lakh to 22 lakh. Deeper bore, better VFD, higher panel capacity, and stronger structure can increase the cost.

4. Can solar run an existing tube well?

Yes, many existing diesel or electric tube wells can be converted to solar. But the pump, motor, bore depth, and water requirement must be checked first.

5. Do solar tube wells need batteries?

Usually no. Agricultural solar tube wells normally run during sunlight hours without batteries. Batteries are expensive for large pumps and are not practical in most farm cases.

6. Which is better: AC or DC solar pump?

For small systems, DC pumps can be useful. For larger agricultural tube wells, AC pumps with VFD are often more practical because of availability, serviceability, and flexibility.

7. How many solar panels are required for a 10HP

pump?

A 10HP pump may need around 10kW to 14kW solar capacity depending on pump load and design. The exact number of panels depends on panel wattage and VFD design.

8. Is solar tube well better than diesel?

For long-term regular use, solar is usually better because running cost is much lower. Diesel may still be useful as backup or for night operation.

9. Can solar tube well work in cloudy weather?

Yes, but output reduces. The pump may run slower or produce less water when sunlight is weak.

10. What is the biggest mistake when buying a solar tube well?

The biggest mistake is buying only by lowest price without checking bore depth, pump sizing, VFD quality, wiring, structure, and after-sales support.

Final Conclusion

The solar tube well price in Pakistan depends on much more than horsepower. A proper system is designed according to your water depth, land size, crop requirement, pump type, solar capacity, VFD, structure, and installation quality.

A cheap system may save money today but create problems tomorrow. A well-designed system gives reliable water, lower running cost, better crop planning, and long-term peace of mind.

If you are serious about installing a solar tube well, do not start with the question, "What is the lowest price?"

Start with the right question:

What system does my land actually need?

Once that answer is clear, the price becomes easier to understand, compare, and justify.

Clear CTA

Need a Solar Tube Well for Your Farm?

FJ Group can help you plan the right solar tube well system according to your land, bore depth, water requirement, and budget.

Whether you want to convert an existing diesel tube well to solar, install a new solar tube well, improve bore water flow, or plan a complete farm water management system, our team can guide you with a practical and professional approach.

Contact FJ Group today for solar tube well consultation, site assessment, and project quotation.

Website: www.fjgroup.pk Phone: 03 111 777 286

FJ Group — Solar, Water, Drilling & Irrigation Project Solutions